mhs-poisson-ltau: implementation

Technical notes on `run.mode = "mhs-poisson-ltau"`. HSEMHS, 24 Oct 2026.

1. Running it

```
[mpiexec -n R] julia --project=. -t T scripts/run.jl config/modes/mhs-poisson-ltau.toml
```

Driver `run_mhs_poisson_ltau` (`src/RunModes.jl`); solver `PoissonTau.solve_tau` (`src/PoissonTau.jl`); MPI layer `src/Slab.jl`; PCG and threaded vector updates from `PoissonZ`. Input and `[tau_grid]` keys as `column-hse-ltau`; in addition:

Key	Default	Meaning
`poisson.logtau_cut`	-4	solved levels $\log\tau \geq$ cut (nearest level k_c)
`poisson.max_steps`	100	Levenberg–Marquardt steps
`poisson.rtol`, `n_consec`	10^{-4}, 3	stop on $\Delta\Phi/\Phi <$ rtol for `n_consec` consecutive steps…
`poisson.dz_tol_m`	100	…or on one such step that moved no column height by more than this
`poisson.cg_tol`, `cg_maxit`	10^{-3}, 80	PCG per trial
`poisson.dc_cap_m`	3×10^5	largest height change of a column per step (whole-step scaling)
`poisson.lm_mu0`	0	initial damping

2. Flow

`column_hse_tau` (aligned, gauged) gives $u = \ln P$ and z_0 on all levels; `solve_tau` solves for u on levels $k > k_c$ and $c = z_{k_c}$ per column, with $u_{k\leq k_c}$ and $z_k - z_{k_c}$ ($k < k_c$) fixed; the heights are gauged (mean $z(\tau{=}1) = 0$); output in the `column-hse-ltau` layout.

3. State, heights and residuals (`evaluate!`)

Nodes $m = 1\ldots n_1$ are levels $k = k_c\ldots n_\tau$. The unknown vector per column has n_1 slots: slot $1 = c$, slots $2\ldots n_1 = u$ at $k > k_c$ (one array (n_1, N_y, n_x), so FFTs and halos treat it uniformly). At each node the EOS gives $\ln\rho$, $\partial\ln\rho/\partial\ln P$ and $\ln\chi$; $\partial\ln\chi/\partial\ln P$ by a central difference ($\varepsilon = 10^{-4}$, the table returns no opacity derivative). With $a = \tau/\chi$, $q = \rho/P$:

$$z_{k+1} = z_k - \tfrac{1}{2}\Delta\ln\tau_k\,(a_k + a_{k+1}), \qquad z_{k_c} = c.$$

Lorentz force at all levels (`lorentz_tau!`): for each component f of $\mathbf{B}$, $\partial_z f = (f_{k-1} - f_{k+1})/(z_{k-1} - z_{k+1})$ (one-sided at the ends), $\partial_x f|_z = D_x f - D_x z\,\partial_z f$, likewise y; $\mathbf{J} = \nabla\times\mathbf{B}/\mu_0$; $\mathbf{L} = \mathbf{J}\times\mathbf{B}$. Residuals with $W = gq - L_z e^{-u}$, $w = Wa$:

$$R_x = -D_x u - W\,D_x z + L_x e^{-u}, \qquad R_{z,f} = u_{f+1} - u_f - \tfrac{1}{2}\Delta\ln\tau_f\,(w_f + w_{f+1}).$$

Weights from the start state (fixed): $\bar H_m$ global median of $1/(gq)$, $\overline{\Delta z}_f$ global median of the level thickness, $w_c = \overline{\Delta z}_{\rm node}\bar H^2$, $w_f = \bar H_f\bar H_{f+1}/\overline{\Delta z}_f$; $\Phi = \sum w_c(R_x^2 + R_y^2) + \sum w_f R_z^2$.

4. Jacobian

Exact, including $\mathbf{L}$'s dependence on the heights (`Jop!`, adjoint `JTop!`):

- **Triangular operator.** $\delta z_m = \delta c + \sum_{f<m}\tfrac{1}{2}\Delta\ln\tau_f\,(a\chi_u)_f\delta u_f + (a\chi_u)_{f+1}\delta u_{f+1}$ with $\chi_u = \partial\ln\chi/\partial\ln P$: a cumulative sum down each column. Its adjoint is a reverse cumulative sum.
- Vertical: $\delta R_{z,f} = \delta u_{f+1} - \delta u_f - \tfrac{1}{2}\Delta\ln\tau_f(w'_f\delta u_f + w'_{f+1}\delta u_{f+1})$, $w' = \partial w/\partial u = W'a - Wa\chi_u$, $W' = gq(\rho_u - 1) + L_z e^{-u}$.
- Horizontal: $\delta R_x = -D_x\delta u - W\,D_x\delta z + \ell_x\delta u$, $\ell_x = -D_x z\,W' - L_x e^{-u}$.
- **Lorentz through the heights.** With $\zeta_m = (\delta z_{m-1} - \delta z_{m+1})/(z_{m-1} - z_{m+1})$ (the relative change of the vertical spacing; δz above the cut $= \delta c$): $\delta(\partial_z f) = -\partial_z f\,\zeta$, $\delta(\partial_x f|_z) = -\partial_z f\,D_x\delta z + D_x z\,\partial_z f\,\zeta$. Hence $\delta\mathbf{J} = \mathbf{J}_\zeta\zeta + \mathbf{J}_X D_x\delta z + \mathbf{J}_Y D_y\delta z$ with $\mathbf{J}_\zeta = (z_y B_{z,z} + B_{y,z},\ -B_{x,z} - z_x B_{z,z},\ z_x B_{y,z} - z_y B_{x,z})/\mu_0$, $\mathbf{J}_X = (0, B_{z,z}, -B_{y,z})/\mu_0$, $\mathbf{J}_Y = (-B_{z,z}, 0, B_{x,z})/\mu_0$, and $\delta\mathbf{L} = \delta\mathbf{J}\times\mathbf{B}$. It enters δR_x through $e^{-u}(\delta L_x + z_x\delta L_z)$ and δR_z through $e^{-u}a\,\delta L_z$: nine node-local coefficient arrays.
- Checks (`MHS_CHECK` in the script version): adjoint $\langle Jv, r\rangle = \langle v, J^\top r\rangle$ to 4×10^{-16}; Jv against central differences of $\mathbf{R}$ to 10^{-9} (with $\mathbf{B}$), 10^{-10} (without). With $\mathbf{L}$ held fixed the directional error was 100 % in magnetic regions and the solve stalled after one step.

5. Linear solve and preconditioner

Normal equations $(J^\top W J + E + \mu D)\,\delta = -J^\top W \mathbf{R}$ by PCG with global inner products; $E = 10^{-2}\kappa^2 D$, $D = (\sum w_c \bar{W}^2,\ w_{c,2}, \dots)$ (the damping shape; slot 1 for c, which no vertical equation sees).

Preconditioner (`setprec!`, `Minv!`): the operator with layer-median coefficients ($\overline{a\chi_u}$, $\overline{w'}$, $\bar{W}$) is, per horizontal mode, $M(s^2) = A + s^2 B$ with s^2 as in `mhs-poisson-z`; $A = G_z^\top W_f G_z + E + \mu D$ (vertical rows), $B = G_h^\top W_c G_h$, $G_h = -(I + \bar{W}\,S)$ the horizontal symbol divided by ik including the triangular matrix S. Both are dense $n_1 \times n_1$ ($n_1 = 55$). Once per step: Cholesky $A = LL^\top$, eigen-decomposition $L^{-1}BL^{-\top} = V\Lambda V^\top$, $Q = V^\top L^{-1}$; then $M(s^2)^{-1} = Q^\top \operatorname{diag}(1/(1 + s^2\lambda))\,Q$ for every mode: two complex GEMMs over all local modes of the transposed FFT layout (`SlabFFT`, as in `mhs-poisson-z`), BLAS threaded during the solve.

6. Steps

Levenberg–Marquardt: for μ (start `lm_mu0`) solve, scale the *whole* step by $\min(1,\ 2/\max|\delta u|,\ \texttt{dc_cap}/\max|\delta c|)$, try up to 4 halvings on the true Φ (EOS status checked); if none decreases Φ, $\mu \leftarrow \max(4\mu, 1)$ and re-solve (up to 8 times). After acceptance $\mu/3$ if the full scaled step was taken, unchanged if one halving, $\times 2$ otherwise; $\mu < 10^{-3} \to 0$. Element-wise clamping of δc (the first version) turned the direction into an ascent direction where the step asked for > 1 Mm in a few columns: Φ rose linearly with the step length. Stop rule as in the table; it replaced a single-step test that stopped one run after 17 steps while strong-field columns still moved 80 km per step.

7. Parallelisation

x slabs; halo exchange in x for every D_x (heights, u, $\mathbf{B}$, and in the Jacobian); global sums; exact global medians for the weights and the preconditioner (bisection on order-preserving integer keys); the distributed FFT (rfft in y, `Alltoallv` transpose, FFT in x) and GEMMs on the local k_y slab. Threads in all node loops.

8. Verification and performance

- 96^2 magnetic crop, ranks $\times$ threads 1×8, 1×32, 4×8: 37 steps each; $z(\tau{=}1)$ differences median 1 m, 99th percentile 12–24 m, max 150 m (weakly determined strong-field columns). The port matches the script `scripts/poisson/poisson_tau_direct.jl`.
- Full cube ($504 \times 504 \times 95$, 54 solved $\ln P$ levels + 1 height per column, 1.4×10^7 unknowns): one node, 96 threads, 17 steps, 14 min, 11 GB; 8 ranks on 4 nodes, 8.3 min, 3.1 GB per rank, same result. PCG reaches its 80-iteration cap in every step (relative residual 10^{-2}–2×10^{-1}): the next speed-up is a better preconditioner.

9. Result on the production cube

$z(\tau{=}1)$ relative to the cube mean, median [5 %, 95 %] in km, by $|\mathbf{B}|$ at $\tau = 1$ (fixed-point = the superseded scheme):

$\lvert\mathbf{B}\rvert$ at $\tau = 1$	column HSE in τ	fixed-point scheme	**mhs-poisson-ltau**
0–200 G	−38 [−85, 187]	−55 [−257, 380]	−10 [−35, 73]
200–500 G	−68 [−88, 58]	+120 [−180, 549]	−26 [−52, 24]
500–1000 G	−86 [−104, −36]	+183 [−198, 587]	−51 [−89, −17]
1000–1500 G	−121 [−144, −92]	+314 [−117, 563]	−96 [−125, −63]
1500–3000 G	−156 [−175, −140]	+402 [58, 568]	−130 [−167, −104]

Force imbalance after the solve, RMS in units of ρg: 0.04–0.09 horizontal and 0.04–0.06 vertical for $\log\tau < 0$ (start 0.3–0.8).

10. Design history

1. Fixed point (`scripts/poisson/poisson_tau.jl`): map to a height grid, Poisson solve there, re-integrate the τ columns, repeat. Each pass is a full Poisson solve (1 h on the full cube), and it diverged where $\mathbf{L}$ carries the gas: the heights were integrated with a pressure that was then replaced. A corrected variant (heights integrated with the Poisson pressure) settled but lifted the strong-field columns instead of depressing them (median +400 km above the cube mean for 1500–3000 G).
2. Direct solve with $\mathbf{L}$ fixed in the Jacobian: exact without $\mathbf{B}$, stalled in magnetic regions.
3. Exact Jacobian + LM with whole-step scaling + the consecutive-step stop rule: the present mode.